



SC2207/CZ2007

Introduction to Databases

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About me

- Instructor (me): Cong Gao
- Prof at CCDS
- Specialized in data management, data mining, Data+AI systems
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- **Emails: Pls put database as part of email title**

Schedule after Recess Week

SQL

- Week 8
- Week 9
- Week 10
- Week 11

Semi-Structured Data
NOSQL/Database topics (not
examinable)

- Week 12

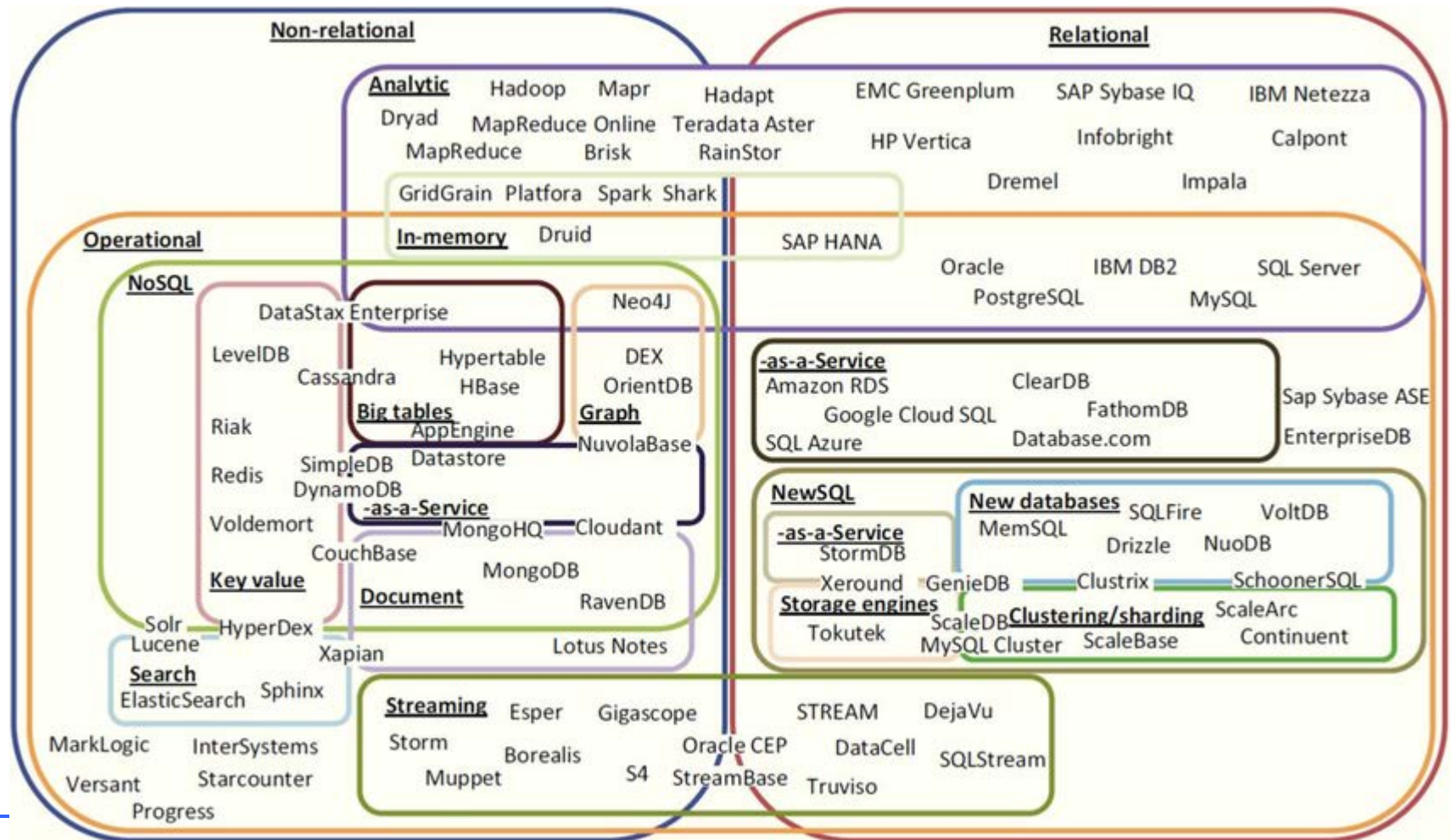
- Week 13 for revision

Big Data Era



Source: dzone.com

Data Systems Landscape



Main Data Systems

Databases and storage

Storage



Data lake



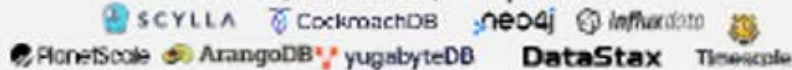
Data warehouse



Vector databases



NoSQL / NewSQL / Graph DBs



DB abstraction



Ingestion, Syncing, and Transport

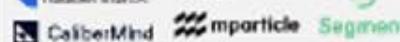
Data ingestion



Reverse ETL



CDP



Querying & processing

Query Engine



Stream processing



Real-time Analytics DB



Core questions related to a data system

1. How does a data system **model** the data?
2. How to **interact** with a data system?
3. How does a data system **store** the data?
4. How does a data system **index** the data?
5. How does a data system answer a **query**?
6. How does a data system support **analytics**?

Plan

■ Lectures

<https://ntu-sg.zoom.us/j/84371949872?pwd=x4cfebLcKPz3FDml0FBzTvlNUtuXam.1>

Meeting ID: 843 7194 9872

Passcode: 123456

■ Tutorial

- Answer will be released at the end of each week
- You should prepare for questions and attend tutorial actively.

■ Lab: please follow your lab tutors.

■ Q & A:

- Zoom session
 - Offline QA: Discussion board in course website. You are encouraged to use it.
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